



Q1. How do we decide on the bandwidth of an oscilloscope? My company wants to buy some oscilloscopes for a new project.

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Through e-mail

A1. Oscilloscope's bandwidth determines the maximum frequency signal that it can measure accurately. The accuracy decreases with increase in signal's frequency. The bandwidth mentioned in the data-sheet (say, 100MHz) is actually the frequency at which a sinusoidal input is attenuated to 70.7 per cent of its true amplitude. Beyond this frequency the oscilloscope cannot support reasonable accuracy.

To decide how much bandwidth you need, find out the range of frequencies you would need to measure. Once you know the range, just use the Five Times rule. Multiply the maximum frequency by five and you have the bandwidth that you would need to accurately measure those signals. For example, to measure signals up to 10MHz frequency accurately, you would need an oscilloscope with 50MHz bandwidth. Otherwise, the high frequency changes will not be resolved and the amplitude will be distorted.

In case of digital signals, it is important to capture the fundamental, third and fifth harmonics to display accurate results. Therefore the bandwidth of the scope, together with the probe, should be at least five times the frequency of the measured signal.

Q2. I have been learning electronics by trying your circuits at home. The circuit diagrams are so neat and pleasing to eyes. What circuit design software do you use?

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